MIL-T-46598E(MU)
29 September 1976
SUPERSEDING
MIL-T-46598D(MU)
29 September 1972

MILITARY SPECIFICATION

TRACER, PROJECTILE: M13, PARTS FOR, AND LOADING, ASSEMBLING AND PACKAGING

"This specification is approved for use by all Department and Agencies of the Department of Defense"

1. SCOPE

1.1 This specification covers the parts for, and the loading, assembling and packaging, for one type of tracer designated as Tracer, Projectile, M13, Assembly for use in Artillery Ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspecfion by Variables for Percent Defective
MIL-STD-1167	- Ammunition Data Cards
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging: Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1234	- Pyrotechnics: Sampling, Inspection, and Testing
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., PA 19137, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1462) appearing at the end of this document or by letter.

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DRAWINGS

Armament Command

D8860550	- Tracer, Projectile, M13 Assembly
C8836133	- Carton, Packing, Ammunition, For Tracer, XM9 & M13
C8836134	- Box, Packing, Ammunition, For Tracer, XM9 & M13
C8796522	- Marking diagram and sealing for Wooden Packing Boxes
LI-8796550	- Index of Inspection Equipment Lists for Tracer, Projectile, M13 Assembly

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of this Code of Federal Regulations (1967 Edition and revision) available from the Superintendent of Ducments, Government Printing Office, Washington, D.C., 20315. Orders for the above publication, cite 49CFR 71-90 (latest revision).)

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

ASTM Methods - American Society for Testing and Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103).

3. REQUIREMENTS

3.1 General

3.1.1 Tracer and Igniter Composition. - The materials used in the tracer and igniter compositions shall comply with the requirements specified on the drawings and referenced specifications.

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3.1.2 Metal Parts. - The material shall comply with the requirements specified on the applicable drawing and referenced specifications.

3.1.3 Tube. - The material shall comply with the requirements of the applicable drawing and referenced specification.

3.2 Metal Defects.

3.2.1 Cup. - The cup, including the cavity, shall be free from cracks, splits, burrs, cold shuts, pipe, porosity, inclusions, folds, seams and other metal defects.

3.2.2 Cup cavity. - The cup cavity shall be free of scale, fins, burrs, drawmarks, laminations, imbedded foreign matter, pits and sharp edges.

3.2.3 Visual Standards. - Visual standards for metal defects will be established for each contract after the first month's production. However, the establishment of visual standards will not waive the requirement for compliance with 3.2.1 and 3.2.2. (See 6.2).

3.3 Workmanship. - The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.3.1 Surface finish. - The requirements for surface finish are as detailed on the applicable drawing.

3.3.2 Burr. - No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.3.3 Foreign matter. - No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention shall be given to the cavity of the body assembly, to insure that no grit, scale, or other foreign matter remains therein after final cleaning.

3.4 Protective coating. - Protective coating shall be in accordance with the applicable drawing and the specifications referenced thereon.

3.5 Moisture Content.

3.5.1 Tracer Composition. - The moisture content of the tracer composition at the loading station at the time of loading of the tracer composition shall not exceed 0.10% when tested as specified in 4.3.3.1 and 4.4.1.

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3.5.2 Igniter Composition. - The moisture content of the igniter composition at the loading station at the time of loading the igniter composition shall not exceed 0.14% when tested as specified in 4.3.3.1 and 4.4.1.

3.6 Tracers. - The tracers shall comply with all the requirements of Drawing D8860550 and with all the requirements specified in the applicable specifications. The tracer shall be free of dirt, chips, cracks, grease, rust and other foreign matter. The cleaning method should not be injurious to the metal nor shall the tracer be contaminated by the cleaning agents used.

3.7 Proving Ground Test. - The tracers shall ignite and there shall be visible evidence of burning of the tracer composition of 2.00 seconds minimum at ambient temperature, 125 degrees Farenheit (+125°F) and minus 40 degrees Farenheit (-40°F) when tested as specified in 4.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions. - Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection. - Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground functional test. (see 3.7).

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the Proving Ground functional tests.

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4.2. First Article Sample.

4.2.1 Initial Production Sample. - At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of assemblies and parts as follows:

55 Tracer Assemblies, complete, including protective coating
5 Cups, including protective coating
5 Cups, uncoated
5 Tubes

Fifty-five of the above complete Tracer Assemblies shall be forwarded to a designated proving ground for burn time testing. The samples shall be manufactured using the same materials, equipment, processes and procedures, including packaging and packing, as will be used in regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacture processes, material used, drawing, specification or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test. - After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure. - Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3. Inspection.

4.3.1 Lot.

4.3.1.1 Submission of Product. - The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Tracer Assembly Lot. - A tracer assembly lot shall consist of tracer assemblies produced by one manufacturer, under one contract, in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision.

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4.3.2 Examination. - Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used in approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Tube, Drawing B8860720, covering a detail of Drawing D8860550

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Inside diameter	Gage
202.	Length	Gage
203.	Wax missing	Visual

4.3.2.2 Cup, Tracer, (prior to applying compound) Drawing B8822518, covering a detail of Drawing D8860550

CRITICAL		
None defined		
MAJOR		
101.	Outside diameter, min.	Gage
102.	Length, min.	Gage
MINOR		
201.	Inside diameter	Gage
202.	Thickness thru bottom	Gage
203.	Burrs	Visual
204.	Foreign matter	Visual
205.	Crack, puncture or other damage	Visual

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4.3.2.3 Cup, Tracer, (prior to insertion of tube) Drawing B8822518, covering a detail of Drawing D8860550.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Coating compound missing, or not completely covering interior surface of cup	Visual
MINOR	
None defined	

4.3.2.4 Cup, Tracer (prior to loading) see Drawing D8860550

CRITICAL	
None defined	
MAJOR	
101. Tube missing	Visual
MINOR	
None defined	

4.3.2.5 Tracer Assembly, Drawing D8860550

CRITICAL	
None defined	
MAJOR	
101. Outside diameter, max.	Gage
102. Total length, max.	Gage
103. Charge missing or less than min, required depth from shoulder	Visual
104. Ignitor charge chipped or broken on any one step in excess of 1/3 of the total area of the step, or total aggregate of chipped or broken area of all steps in excess of 1/3 of the total area of the largest step	Visual
105. No Multiple parts if not required	Visual

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4.3.2.6 Carton, packing, prior to sealing, Drawing C8836133

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Any packing material missing	Visual
202. Tracer missing	Visual

4.3.2.7 Carton, packing, sealing, Drawing C8836133

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Sealing strip fails to completely seal carton	Visual
202. Carton torn, cut or punctured	Visual
203. Marking missing, misleading or unidentifiable	Visual

4.3.2.8 Sealed wood packing box, Drawing C8837866

CRITICAL	
None defined	
MAJOR	
101. Box damaged exposing contents	Visual
102. Board loose or protruding nail	Visual
103. Strapping broken, missing or loose	Visual
104. ICC nomenclature missing, misleading or unidentifiable	Visual
105. Cover improperly assembled	Visual
106. Contents loose	Manual

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Strapping improperly engaged	Visual/Manual
202. Marking missing, misleading, or unidentifiable	Visual

4.3.3 Testing

4.3.3.1 Moisture and total volatile content of tracer composition and igniter composition. - The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test a minimum of one randomly selected sample from each batch or from each 8 hours production, whichever is the lesser amount. Composite samples shall not be used for determining compliance. Failure to comply with the applicable requirement shall be cause for rejection of the parent quantity represented by the sample. Testing shall be performed as specified in 4.4.1.

4.3.3.2 Proving Ground.

4.3.3.2.1 Functioning (see 3.7). - This test shall be performed as specified in 4.4.3.1.

4.3.3.2.1.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B-3, Code letter I and AQL 1.50 percent. If any tracer fails to ignite or fails to produce visible evidence of burning, the lot shall be rejected.

4.3.3.2.1.2 After three consecutive lots have met the criteria of 4.3.3.2.1.1, the sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B-3, Code letter F and AQL 1.50 percent. If any tracer fails to ignite or fails to ignite or fails to produce visible evidence of burning, the lot shall be rejected.

4.3.3.2.1.3 Refiring. - If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional tracers as required.

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4.3.4 Inspection Equipment. - LI-8860550 identifies the inspection equipment required to perform the examinations and tests prescribed herein.

4.4 Test methods and procedures.

4.4.1 Determination of moisture and total volatile content of tracer composition and igniter composition. - The moisture and total volatile content of the tracer composition and igniter composition shall be determined in accordance with MIL-STD-1234 Method 102.2 (see 6.4).

4.4.2 Protective coating. - The procedure for testing the cleaning of components and the application of protective coating shall be in accordance with the critieria for testing, as specified in the applicable specification. The salt spray test shall be performed utilizing the procedures and equipment prescribed by ASTM Method B-117.

4.4.3 Proving Ground.

4.4.3.1 Ballistic tracer functioning (acceptance test). - The sample tracers shall be assembled to and tested in Cartridge, 105MM, APDS-T, M392 Series or Cartridge, 105MM, TPDS-T, M724. The assembled rounds shall be fired at the temperature listed in Table I from the appropriate weapon and the tracer burn times recorded. A timing device capable of measuring time to the nearest 0.01 second shall be used to measure burning time. Suitable aids, such as binoculars, should be used if necessary to insure visible contact with tracer, (see 6.9 and 6.10).

TABLE I

Temperature	Quantity
Ambient	25
+125°F 1/	15 2/
- 40°F 1/	15 2/

1/ The rounds shall be temperature conditioned for a minimum period of 18 hours.

2/ After the first article sample and three (3) consecutive production lots have been tested successfully, the temperature conditioning phase may be discontinued with the approval of the procuring agency. However, implementation of the temperature conditioning phase shall be instituted whenever directed by the procuring contracting officer (P.C.O.).

4.4.4 Government use of contractor's inspection equipment. - The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 The tracer assemblies shall be preserved and packaged in accordance with Drawing C8836133 and the requirements of Level C in MIL-STD-1169.

5.2 Packing.

5.2.1 The cartons shall be packed in accordance with Drawing C8836134 and the requirements of Level C in MIL-STD-1169.

5.3 Marking.

5.3.1 The cartons and boxes shall be marked in accordance with Drawings C8836133, C8836134 and MIL-STD-1168.

5.4 Data Card. - Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification. - When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards. - Visual standards to be established at contractor's plant in accordance with 3.2.3.

6.3 Ordering data. - Procurement documents should specify the title, number, and date of this specification.

6.4 Moisture. - The igniter and tracer compositions, and the pellets form thereof, are hygorscopic and will deteriorate rapidly when exposed to moisture. Small smounts of moisture present in or absorbed by the compositions or pellets, prior to or after loading into tracers will eventually result in unsatisfactory functioning. It is highly desirable that the ingredients, compositions, pellets, and loaded tracers be carefully safeguarded from exposure to moisture and humid atmospheres. For assurance of a satisfactory loaded tracer, it is recommended that the blending and pelletizing buildings be air conditioned in a manner which will hold the humidity of the atmosphere as low as practicable.

6.5 Alternative testing procedures. - Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions psecified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for approval of the responsible technical agency that their affectiveness is equal to or better than the quality assurance provisions specified herein.

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6.6 When a product is being inspected in accordance with MIL-STD-1235 and a defective characteristic is found, the product produced prior to the discovered defect will be screened back 2000 tracer assemblies and all defects culled out.

6.7 Shipping. - It is not intended to ship these compositions or pellets. It is intended that the compositions and pellets thereof be prepared at the plant where the loading of the tracers will be accomplished. If shipment is necessary, Department of Transportation (DOT) Class and Marking cited on the affected item drawings is applicable. Material must be adequately and properly packaged in accordance with applicable DOT Packaging Requirements (Para 173.61, CODE OF FEDERAL REGULATIONS (CFR) Title 49.).

6.8 Hazard notice. - The tracer described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacutre, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

6.9 Combining of proving ground tests. - When the contractor for the tracer is also contractor for the complete round or one of the loaded components thereof, the proving ground tests may be combined in order to save expense, upon agreement between the procuring activity and the contractor.

6.10 Submission of firing and test data. - In addition to the normal distribution of records for each lot of tracers, when the item is procured by the Department of Defense, copies should be sent to:

Commander
Frankford Arsenal
ATTN: SARFA-QAA-Q(1) (1 copy)
SARFA-MDP-Y (1 copy)
Philadelphia, PA 19137

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project Number 1315 - A287

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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